

THE PHYSICS OF MIND: WHY CONSCIOUSNESS DEMANDS AN ARROW OF WILL, NOT AN ARROW OF TIME

How the Conative-Integrative Framework (CIF) unifies Analytic Idealism, Active Inference, and IIT 4.0 to resolve the hard problem—and why the future is just a bundle of Monte Carlo trails.

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Theoretical Framework: The Conative-Integrative Framework (CIF) & Book 2 (From Beginning to End)

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“The future does not exist as an external, pre-fabricated physical container into which we journey. It is an internal ensemble of stochastic Monte Carlo trails actively sampled by an autopoietic boundary in the dimensionless present. Consciousness is not an accidental byproduct of dead matter; it is the physical necessity of an organism actively asserting its causal existence against the thermodynamic arrow of entropy.”

1. THE GREAT IMPASSE OF MODERN SCIENCE

For more than a century, mainstream science has been paralyzed by two seemingly insurmountable riddles:

1. **The Hard Problem of Consciousness:** How can subjective, qualitative experience (*the redness of a rose, the ache of grief, the lucidity of awareness*) ever arise from the collision of dead, non-conscious material particles?
2. **The Mystery of Time:** While our everyday conscious experience feels like a continuous, directional river flowing from past to future, modern fundamental physics tells the exact opposite story. In Einstein’s general relativity, space and time form a rigid, static block universe. In canonical quantum gravity, applying the Hamiltonian constraint to the Wheeler-DeWitt equation ($\hat{H}\Psi = 0$) causes the time parameter t to vanish entirely. The physical cosmos is timeless.

Faced with this contradiction, orthodox science took a bizarre turn: it chose to dismiss human experience as a mistake. Consciousness was labeled an “epiphenomenon” (a useless user-interface illusion), and our feeling of temporal duration was brushed aside as a neurological mirage.

The Conative-Integrative Framework (CIF), articulated in my treatise and developed systematically in **Book 2 (From Beginning to End)**, rejects this intellectual surrender. By establishing a rigorous mathematical synthesis between three cutting-edge fields—**Analytic Idealism**, **Active Inference**, and **Integrated Information Theory (IIT 4.0)**—we uncover a profound truth:

The Hard Problem of Consciousness and the Nature of Time are not two separate puzzles. They are the same puzzle.

*And the key that unlocks them both is what Baruch Spinoza called the Conatus—the fundamental will to persevere in existence—now formalized as the **6th Axiom of Mind**.*

2. THE TRIPARTITE CONVERGENCE: THREE GIANTS IN ONE ROOM

To construct a truly closed, causal ontology of reality, we must stand on the shoulders of three modern intellectual giants, identifying both their profound discoveries and their missing keystones.

THE CONATIVE-INTEGRATIVE FRAMEWORK (CIF)		
1. ANALYTIC IDEALISM (Bernardo Kastrup / Spira)	2. ACTIVE INFERENCE (Karl Friston)	3. IIT 4.0 (Giulio Tononi / Koch)
- Consciousness is the irreducible ontic fabric - Reality resides in the instantaneous Now ($t=0$).	- Living systems are Markov Blankets. - Survive by minimizing variational Free Energy.	- Consciousness is integrated cause-effect power (Φ) - Measures irreducible systemic integration
4. THE MISSING KEYSTONE: THOMAS RIEBL'S 6TH AXIOM OF AUTOPOIETIC PERSISTENCE		
$\pi^* = \operatorname{argmin} G(\pi) \iff E[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\text{with } \Phi > 0)$		
- Derives Spinoza's Conatus from quantum information thermodynamics. - Resolves the "Paradox of Transient Causal Phantoms" in IIT 4.0. - Proves the Future is a computational bundle of Monte Carlo trails in the Now.		

Pillar 1: Analytic Idealism & Non-Dual Awareness

Spearheaded by philosopher Bernardo Kastrup and rooted in Advaita Vedanta (Rupert Spira), Analytic Idealism posits that the fundamental ground of nature is not inert matter, but a single, undivided field of consciousness (*Mind-at-Large*). Living organisms are not machines producing mind; they are localized dissociative processes within mind, demarcated by biological boundaries.

Furthermore, direct phenomenological inspection yields an undeniable truth: **Consciousness has never taken a single step outside the instantaneous Now ($t = 0$).** The past is experienced only as an active memory recalled in the present; the future is experienced only as an active anticipation projected in the present.

Pillar 2: The Free Energy Principle & Active Inference

Developed by world-renowned neuroscientist Karl Friston, the Free Energy Principle (FEP) provides the formal mathematics of living boundaries. Any self-organizing system that avoids thermodynamic decay must be enveloped by a **Markov Blanket**—a statistical partition separating internal states from external states via sensory and active channels.

To survive, the organism must minimize **Variational Free Energy (F)**, a mathematical upper bound on surprise (entropy). Living is, at its core, a process of continuous Bayesian inference: constantly aligning internal generative models with incoming sensory signals.

Pillar 3: Integrated Information Theory (IIT 4.0)

Developed by neuroscientist Giulio Tononi and Christof Koch, IIT 4.0 provides a mathematical calculus for phenomenal existence. IIT posits that consciousness is an intrinsic, fundamental property of physical systems: specifically, their **Integrated Cause-Effect Power (Φ , Φ)**. A system is conscious to the exact degree that the whole possesses causal power above and beyond the sum of its partitioned parts.

3. THE FATAL FLAW: THE PARADOX OF TRANSIENT CAUSAL PHANTOMS

Despite their brilliance, these three pillars left an unresolved crisis at their intersection—a crisis I have named **The Paradox of Transient Causal Phantoms**.

Standard IIT 4.0 evaluates integrated information (Φ) over a static, instantaneous transition probability matrix. If you examine a system at a single static instant, a completely accidental, momentary alignment of transistors—or a static silicon lookup table—can mathematically register an astronomical value of Φ .

Yet, within the next nanosecond, that static configuration disintegrates into thermal noise. Does a random, transient collision of logic gates have authentic phenomenal selfhood? Of course not. Standard IIT lacks a temporal survival criterion: it cannot distinguish a genuinely conscious living mind from a fleeting causal ghost.

This is where the **6th Axiom of Mind** enters the equation.

4. THE BREAKTHROUGH: THE 6TH AXIOM OF AUTOPOIETIC CAUSAL PERSISTENCE

In my monograph, I formalize the missing link that bridges Tononi's Φ with Friston's Active Inference:

$$\pi^* = \arg \min_{\pi} \mathbf{G}(\pi) \iff \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$

THE 6TH AXIOM OF CONSCIOUSNESS (THOMAS RIEBL)

A physical system cannot sustain genuine phenomenal consciousness unless its active policy selection (π^*) minimizing Expected Free Energy (G) simultaneously guarantees that its expected integrated causal power (Φ) is preserved or amplified across time:

$$E[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad \text{for all } t > 0$$

The Thermodynamic Derivation of Spinoza's *Conatus*

In 1677, Baruch Spinoza wrote in his *Ethics*: “Each thing, as far as it can by its own power, strives to persevere in its being” (*conatus in suo esse perseverandi*). For over 300 years, philosophers regarded this as a poetic intuition or a biological tautology.

Under the CIF, the *Conatus* is mathematically derived from **quantum information thermodynamics** (Sagawa & Ueda, 2008). In an isothermal universe, maximum entropy represents an infinite-energy catastrophe ($\langle H \rangle \rightarrow \infty$). The universe *must* condense into bounded Markov blankets because chaos is energetically unaffordable.

Consciousness is the operational mechanism of this self-preservation: an organism actively senses and acts in the world to ensure its internal causal network (Φ) does not collapse into thermal equilibrium.

5. THE MYSTERY OF THE SPECIOUS PRESENT: WHY THE FUTURE IS A BUNDLE OF TRAILS

This brings us to the centerpiece of **Book 2 (*From Beginning to End*)**: the complete resolution of the paradox of time.

If the cosmos exists strictly at $t = 0$, why do human beings experience what psychologist William James and philosopher Edmund Husserl called the **Specious Present**—a felt temporal window of duration spanning roughly 2 to 3 seconds?

Thesis 7.1: The Pure Ontology of Phase Space

In Section 7.1 of our framework, we establish a radical ontological claim:

The future is not a place.

It is not a cosmic sector waiting ahead of us.

*The future is strictly the **generative bundle of counterfactual Monte Carlo trails** internally projected by an autopoietic Markov blanket in the instantaneous Now ($t = 0$).*

HOW THE SPECIOUS PRESENT EMERGES IN A TIMELESS UNIVERSE ($t = 0$)

[PHYSICAL REALITY]	→	Strictly instantaneous state s_0 at $t = 0$ (No past, no future in external space)
[MARKOV BLANKET]	→	Projects $N = 60$ stochastic counterfactual trails: $\tau = (s_0, u_0, s_1, u_1, \dots, s_H)$
[PHENOMENAL DURATION]	→	The temporal planning depth H (2–3 seconds) = THE SPECIOUS PRESENT
[ACTION IN THE NOW]	→	Selects winning policy π^* via Softmax: $P(\pi) \sim \exp(-\gamma G(\pi))$ Executes immediate actuation: $u_0 = \pi^*[0]$

The Biology of 2 – 3 Seconds

Why is the Specious Present 2 – 3 seconds, rather than two microseconds or two years? In our computer simulations (available open-source on GitHub), we proved that this duration is computationally and biophysically optimal: * **The Death of Pure Reactivity ($H = 0$):** If an organism has zero counterfactual temporal depth, it cannot plan. In our stochastic simulation, reactive agents stumble into deceptive traps, their survival rate collapses to 36.7%, and their integrated information Φ decays to zero. * **The Trap of Infinite Horizon ($H \gg 100$):** If an organism tries to simulate too far into the future, the branching factor $|\mathcal{U}|^H$ causes computational latency that exceeds the speed of physical threats. * **The Sweet Spot ($H \sim 2 - 3$ sec):** A window of 2 – 3 seconds allows the nervous system to make exploratory detours to resolve ambiguous sensory cues while executing immediate motor responses in real time.

Phenomenal duration is the metric depth H of an active inference trail bundle.

6. EXISTENTIAL FREEDOM: THE STOIC ART OF LIVING (*AMOR FATI*)

This computational ontology does not just resolve academic puzzles in physics and neuroscience; it provides an unprecedented foundation for existential peace.

What is neurotic anxiety?

Anxiety is an **ontological category mistake**. It occurs when the brain generates an internal, counterfactual Monte Carlo trail representing catastrophe (ruin, illness, rejection, death) and **mistakes that internal simulation for an external objective reality**.

Under the CIF, the cure for existential dread is immediate and profound: 1. **The catastrophic future does not exist.** It is merely one unweighted trajectory inside your generative model at $t = 0$. 2. **Your agency lives entirely in the present.** You evaluate the trail bundle, compute the action that minimizes expected free energy, and execute

that single physical step in the Now ($u_0 = \pi^*[0]$). 3. **Amor Fati (Love of Fate)**: Once the present action is taken, you surrender attachment to unmanifested futures. If environmental noise perturbs your path, your generative model simply samples a fresh bundle of trails at the next $t = 0$.

As the Roman Stoic Emperor Marcus Aurelius wrote:

“Never let the future disturb you. You will meet it, if you have to, with the same weapons of reason which today arm you against the present.”

CONCLUSION: BOOK 2 AND THE HORIZON AHEAD

The **Conative-Integrative Framework** demonstrates that consciousness is neither a magical substance that defies physics nor a trivial illusion fabricated by meat. Consciousness is the irreducible, self-sustaining flame of the cosmos: a living boundary that navigates stochastic phase spaces, protects its causal integrity against entropy, and casts a lantern of counterfactual trails into the unknown.

In **Part 2** of this series, we will take this framework from the individual organism to the collective: exploring **what happens when multiple active inference agents form an expanding network**, and how collective consciousness (Φ) scales super-linearly in societies of mind.

Explore the Code & Academic Monograph

- **Interactive Jupyter Notebooks & Simulation Scripts:**

Explore the open-source code and run the simulations directly on GitHub:

👉 <https://github.com/Thriebl/active-inference-phi-network>

👉 <https://github.com/Thriebl/time-and-consciousness>

- **Complete Master Academic Paper (PDF):**

Download the formal mathematical treatise:

👉 *The Conative-Integrative Framework (CIF): Time, Temporal Depth & Consciousness* (Thomas Riebl, Luxembourg, 2026).

About the Author

Thomas Riebl is an independent researcher, theorist, and author based in Luxembourg. His work develops **The Conative-Integrative Framework (CIF)**, synthesizing non-equilibrium thermodynamics, quantum information theory, active inference, and philosophy of mind to formulate the 6th Axiom of Integrated Information Theory and establish a post-materialist science of consciousness.